

Research Interests

Human-Computer Interaction, focusing on AI-powered input technologies, accessibility, and immersive VR/AR interactions; developing LLM-based methods to improve user performance, experience, and inclusivity.

Work Experience

2025–Present **Assistant Professor**, *Kean University*, Department of Computer Science and Technology.

Education

- 2019–2025 **Ph.D. in Computer Science**, *Stony Brook University*, GPA: 3.98/4.0.
2022 M.S. awarded during Ph.D. program
- 2017–2019 **B.S. in Computer Science**, *Binghamton University*, GPA: 4.0/4.0.
Institutional Honors: Summa Cum Laude (Ranked 1st in Class of 2019)
- 2014–2018 **B.S. in Applied Mathematics**, *Renmin University of China*, Major GPA: 3.58/4.0.
Second-Class Academic Excellence Scholarship (Top 15/133)

Publications

Accepted Articles

- 2026 Tony Li, **Yan Ma**, Zhuojun Li, Chun Yu, IV Ramakrishnan, and Xiaojun Bi. Keysense: Llm-powered hands-down, ten-finger typing on commodity touchscreens. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI '26, New York, NY, USA, 2026. Association for Computing Machinery.
- 2025 Dan Zhang, **Yan Ma**, Glenn Dausch, William H Seiple, Xianfeng David Gu, IV Ramakrishnan, and Xiaojun Bi. Enabling auto-correction on soft braille keyboard. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, 2025.
- 2025 **Yan Ma**, Dan Zhang, IV Ramakrishnan, and Xiaojun Bi. LLM powered text entry decoding and flexible typing on smartphones. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025.
- 2025 **Yan Ma**, Tony Li, Zhi Li, and Xiaojun Bi. LLM-powered text entry in virtual reality. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, 2025.
- 2025 Hyunchul Lim, Nam Anh Dang, Dylan Lee, Tianhong Catherine Yu, Jane Lu, Franklin Mingzhe Li, Yiqi Jin, **Yan Ma**, Xiaojun Bi, Francois Guimbretiere, and Cheng Zhang. Spellring: Recognizing continuous fingerspelling in american sign language using a ring. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025.
- 2023 Jeremy Chu, **Yan Ma**, Shumin Zhai, Xianfeng David Gu, and Xiaojun Bi. Touchtype-gan: Modeling touch typing with generative adversarial network. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.
- 2023 Jeremy Chu, Dongsheng An, **Yan Ma**, Wenzhe Cui, Shumin Zhai, Xianfeng David Gu, and Xiaojun Bi. Wordgesture-gan: modeling word-gesture movement with generative adversarial network. In *Proceedings of the 2023 CHI conference on human factors in computing systems*, 2023. 🏆 **Best Paper Honorable Mention Award**.

- 2022 Zhi Li, Maozheng Zhao, Dibyendu Das, Hang Zhao, **Yan Ma**, Wanyu Liu, Michel Beaudouin-Lafon, Fusheng Wang, Iv Ramakrishnan, and Xiaojun Bi. Select or suggest? reinforcement learning-based method for high-accuracy target selection on touchscreens. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 2022.
- 2022 Sophia Gu*, **Yan Ma***[†], Zhi Li, Xiangmin Fan, Feng Tian, and Xiaojun Bi[†]. Using deep learning to detect motor impairment in early parkinson's disease from touchscreen typing. In *Graphics Interface 2022*, 2022. * **Co-first author**, [†] **co-supervised Sophia**.
- 2021 **Yan Ma**, Shumin Zhai, IV Ramakrishnan, and Xiaojun Bi. Modeling touch point distribution with rotational dual gaussian model. In *The 34th annual acm symposium on user interface software and technology*, 2021.
- 2019 Zilai Li, **Yan Ma**, and Yaobin Ou. Global strong solutions to 1-d vacuum free boundary problem for compressible navier–stokes equations with variable viscosity and thermal conductivity. *Journal of Mathematical Analysis and Applications*, volume 474, pages 1153–1177. Elsevier, 2019.

Teaching Experience

Kean University, Union, NJ

Fall 2025 – **Course Instructor**.

- Present
- **TECH3620: Android Application Development** (Spring 2026)
 - **CPS2231: Computer Programming** (Fall 2025, Spring 2026, Fall 2026)
 - **CPS3440: Analysis of Algorithms** (Fall 2025, Fall 2026)

Stony Brook University, Stony Brook, NY

Fall 2019 – **Teaching Assistant / Guest Lecturer**.

- Spring 2025
- **CSE/ISE/EST323: Human-Computer Interaction** (Guest Lecturer & TA: Spring 2021, Spring 2025)
 - **CSE548: Analysis of Algorithms (Graduate Level)** (Fall 2020)
 - **CSE303: Introduction to the Theory of Computation** (Fall 2019, Spring 2020)
 - Contributed to a problem-solution book (in preparation)

Student Research Mentoring

Kean University, Union, NJ

Spring 2026 **Master's Thesis Committee Member**, Student: Arthur Rodriguez, *Investigating the Influence of Simulated Agents on Pedestrian Road-Crossing Decision-Making Using a Virtual Environment*.

Summer 2026 **CRA UR2PhD REU**, Mentored two undergraduate researchers on *TraceVis*, an AI-assisted interactive Java execution visualization system.

Summer 2026 **Students Partnering with Faculty (SpF)**, Mentored two undergraduate researchers on *Detecting and Staging Parkinson's Disease from Smartphone Interaction Data Using Deep Learning*.

Spring 2026 **CAHSI LREU**, Mentored one undergraduate researcher on *MindMotion*, a smartphone-based application for Parkinson's disease detection.

Spring 2026 **CRA Undergraduate Research Course**, Mentored three undergraduate researchers on *Debate Coach*, an AI-powered tool that challenges users' arguments with fact-based counterarguments to promote critical thinking..

Fall 2025 – **Undergraduate Research Mentoring**, In addition to the funded mentorships listed above, Present mentoring four undergraduate students across four research projects through weekly meeting.

Stony Brook University, Stony Brook, NY

Summer 2024 **Computer Science & Informatics Summer Research Experience (CSIRE) Program Mentor**, Mentored high school student Jerry Yao on a VR text editing project, helping him develop key research skills. Jerry presented his research at the CSIRE seminar as a presentation and a poster.

Summer 2021 **CSIRE Program Mentor**, Mentored high school student Sophia Gu on a Parkinson's disease detection research project, providing guidance that contributed to a published paper.

Spring 2021 **Master's Student Co-Mentoring at Stony Brook University**, Co-mentored a master's student on an independent study project exploring any-surface typing using a depth camera.

Academic Service

- 2026 **Senior Program Committee Member**, AAAI ICWSM.
- 2025–2026 **Program Committee Member**, AAAI ICWSM 2025; ACM IUI 2026.
- 2026 **Conference Reviewer** (21 reviews), ACM CHI, ACM/IEEE HRI, IEEE VR, ACM IUI, ACM SUI, ACM UIST (**Special Recognition for Outstanding Review**), ACM VRST, AAAI ICWSM.
- 2025 **Conference Reviewer** (24 reviews), ACM CHI, ACM DIS, IEEE VR, ACM IMWUT, ACM IUI, ACM MobileHCI, ACM SUI, ACM UIST, ACM UbiComp/ISWC, IEEE VIS, AAAI ICWSM.
- 2024 **Conference Reviewer** (13 reviews), ACM CHI, ACM CSCW, ACM CUI (**Special Recognition for Outstanding Review**), ACM IDC, ACM IMX, ACM ISS, ACM SUI, ACM UIST.
- 2023–2026 **Journal Reviewer**, International Journal of Human–Computer Studies (IJHCS), a Q1 journal; approximately 2 manuscripts annually.
- 2026 **Journal Reviewer**, International Journal of Human–Computer Interaction (IJHCI), a Q1 journal; 1 manuscript annually.

Professional Activities & Engagement

- 2026 **Fourth Kean University Faculty Research Conference**, *Kean University*, Presenter.
- 2025 **Invited Talk: *AI-powered Text Input Technologies***, Department of Electrical Engineering and Computer Science, South Dakota State University.